	R 6.0
6(a)	Magnetic flux density is defined as the force per unit length per unit current acting on a straight conductor placed perpendicularly to the magnetic field.
6(b)	$R = \dfrac{\rho L}{A} = \dfrac{(1.7\times10^{-8})\left[520\times\pi(4.0\times10^{-2})\right]}{\frac{\pi}{4}(0.46\times10^{-3})^2} = 6.7\ \Omega$
6(c)	$B = \mu_0 n I = (4\pi\times10^{-7})\left(\dfrac{520}{(520)(0.46\times10^{-3})}\right)\left(\dfrac{24}{6.68}\right) = 9.8\times10^{-3}\ \text{T}$
6(d)	The magnitude of the force will be zero, as the current in the wire is parallel to the direction of the magnetic flux density.
7(a)	A photon is a quantum of electromagnetic radiation, where the energy in a photon